

Difficulty Breathing - Asthma / COPD

History:	Signs / Symptoms:	Differential Diagnosis:
- Gradual onset is more common - Medication often prescribed – Albuterol, Atrovent, Singulair, steroids - Chronic cough, productive cough - Tobacco use - Exercise or cold induced wheezing	- Respiratory distress - Bilateral wheezes or stridor - Diminished breath sounds - Accessory muscle use - Chest tightness/Cough - Tachycardia	- Anaphylaxis - Aspiration - CHF - Pulmonary edema - Pericarditis - Pneumonia - Pneumothorax - Pulmonary embolus - Myocardial infarction - Toxic inhalation

EMT/AEMT

- Follow General Medical Care Guideline
- Sit upright or in optimal position to minimize respiratory distress
- Assist patient with **Albuterol** metered dose inhaler (*MDI*) - MUST USE SPACER
 - 2-4 inhalations
 - Repeat every 20 minutes for 3 doses
 - Then repeat every 1-2 hours as needed
- Follow Adult Resuscitation Guideline, as indicated

EMT/AEMT - Expanded Scope (see Paramedic section below for indications and doses)

- Albuterol and Ipratropium Bromide
- Epinephrine IM or Nebulizer
- Prednisone PO, Dexamethasone IM or PO, Solumedrol IM

Paramedic

- Albuterol** 2.5mg Nebulized with Ipratropium Bromide 500mcg Nebulized
 - May repeat x 2
 - Additional Albuterol 2.5 mg / 3 mL continuous until symptoms resolve
- Epinephrine 1:1000 0.5mg IM**
 - For severe respiratory distress (tripoding, diminished breath sounds, hypoxia)
 - May repeat every 5-10min until no longer in respiratory distress
- Steroid Therapy**
 - Dexamethasone** 10 mg IV/IO/IM/PO OR
 - Solumedrol** 125 mg IV/IO/IM OR
 - Prednisone** 60mg PO
- Magnesium Sulfate** 2 g in 100 mL NS over 10 min (for asthma only)
- Epinephrine Nebulizer** Treatment 3 mg (3 vials of 1:1000) mixed with 2ml saline
- If assisting ventilations, pay close attention to SpO₂, CO₂ waveform (if available), and resistance to bagging
 - Asthma and COPD are high risk for pneumothorax
 - First give a compress the chest to release air that may be restricted within the lungs
 - If unresolved, utilize pleural decompression techniques

If blood pressure is over 180 systolic, patient had sudden onset of shortness of breath or has signs of fluid retention (edema) and the patient has history of potential history of cardiac disease, consider pulmonary edema as cause of respiratory distress. Avoid sympathomimetics such as epinephrine in these patients.